PharmAchieve

# DIABETES DRUG CHARTS

Qualifying Exam Preparatory Course

# LEGEND

| | |
|---|---|
| **ASCVD** | Atherosclerotic Cardiovascular Disease |
| **BID** | Twice Daily |
| **CKD** | Chronic Kidney Disease |
| **CV** | Cardiovascular |
| **D** | Diarrhea |
| **DKA** | Diabetic Ketoacidosis |
| **eGFR** | Estimated Glomerular Filtration Rate |
| **ESLD** | End-stage Liver Disease |
| **ESRD** | End-stage Renal Disease |
| **ER** | Extended Release |
| **FBG** | Fasting Blood Glucose |
| **GI** | Gastrointestinal |
| **HbA1C** | Hemoglobin A1C |
| **HF** | Heart Failure |
| **HF-rEF** | Heart Failure With Reduced Ejection Fraction |
| **IBD** | Inflammatory Bowel Disease |
| **IR** | Immediate Release |
| **MACE** | Major Atherosclerotic Cardiovascular Events |
| **MAO** | Monoamine Oxidase |
| **MENS-2** | Multiple Endocrine Neoplasia, Type II |
| **MI** | Myocardial Infarction |
| **MTC** | Medullary Thyroid Cancer |
| **MR** | Modified Release |
| **N** | Nausea |
| **PPAR** | Peroxisome Proliferator-activated Receptor |
| **PO** | Orally, By Mouth |
| **PPG** | Postprandial Glucose |
| **RA** | Rapid Acting |
| **RF** | Risk Factor |
| **SA** | Short Acting |
| **SUBQ** | Subcutaneous |
| **TID** | Three Times Daily |
| **TZD** | Thiazolidinediones |
| **V** | Vomiting |

# PHARMACOLOGICAL MEASURES

## BIGUANIDES & α-GLUCOSIDASE INHIBITORS

| Drug class/A1C lowering | MOA | Safety | C/I and D/I |
|---|---|---|---|
| **Biguanides**<br>• Metformin<br><br>~1% | • ↑ peripheral glucose uptake and insulin sensitivity<br>• ↓ hepatic glucose production | • AEs: upset stomach, N/V/D, metallic taste, lactic acidosis,<br>• B12 deficiency<br>• Low risk of hypoglycemia<br>• No weight gain | • C/I: eGFR < 15 ml/min/1.73 $m^2$, severe hepatic failure (including AUD), severe dehydration, hx of lactic acidosis<br>• Lower dose if eGFR < 45 ml/min/1.73 $m^2$; monitor renal fxn and for signs of lactic acidosis<br>• D/I: cimetidine & dolutegravir (↑metformin) |
| **α-Glucosidase Inhibitors**<br>• Acarbose<br><br>~0.7-0.8% | • Inhibit α-glucosidase enzyme in SI<br>• Less glucose is absorbed since carbohydrates are not broken down | • AEs: flatulence, diarrhea, abdominal pain (dose related), hepatitis | • C/I: IBD, intestinal obstruction, intestinal diseases, ulceration, severe hepatic disease<br>• ↓ digoxin levels<br>• ↓ effect when used with digestive enzymes |

# PHARMACOLOGICAL MEASURES

## INSULIN SECRETAGOGUES

| Drug class | MOA | Safety | C/I and D/I |
| --- | --- | --- | --- |
| **Sulfonylureas**<br>• Gliclazide<br>• Glyburide<br><br>~0.6-1.2% | • Depolarizes beta cells to stimulate insulin release | AEs:<br>• Hypoglycemia (higher with glyburide and eGFR < 60 ml/min)<br>• Weight gain (1.2 - 3.2 kg)<br>• Headache<br>• Dizziness<br>• GI effects | • C/I: T1DM, pregnancy, breastfeeding, ketoacidosis (with/without coma), severe liver/renal impairment, repaglinide with clopidogrel or gemfibrozil<br>• Alcohol (↑ hypoglycemia)<br>• β blockers (may mask hypoglycemia with sulfonylureas)<br>• Drugs that cause hyper or hypoglycemia<br>• CYP 3A4 Inhibitors/Inducers (meglitinides only) |
| **Meglitinides**<br>• Repaglinide<br><br>~0.7-1.1% | | AEs:<br>• Hypoglycemia (lower than sulfonylureas)<br>• Weight gain (1.4 - 3.3 kg) | |

# PHARMACOLOGICAL MEASURES

## THIAZOLIDINEDIONES & DPP-4 INHIBITORS

| Drug class | MOA | Safety | C/I and D/I |
|---|---|---|---|
| **Thiazolidinediones**<br>• Pioglitazone<br>• Rosiglitazone<br><br>~0.5-1.5% | • ↑ PPAR-γ sensitivity to insulin<br>• ↓ hepatic glucose production | AEs:<br>• Worsening HF, fluid retention, weight gain (2 – 2.5 kg), fractures, macular edema (rare), URI, headache | • CYP 2C8 inhibitors/inducers (gemfibrozil - prolonged hypoglycemia)<br>• Antihyperglycemic agents<br>• C/I: HF, liver failure, bladder cancer with pioglitazone, MI with rosiglitazone – controversial |
| **DPP-4 Inhibitors**<br>• Sitagliptin<br>• Saxagliptin<br>• Linagliptin<br>• Alogliptin<br><br>~0.5-0.7% | • DPP-4 inhibition leads to ↑ in incretin levels | AEs:<br>• Nasopharyngitis, hypersensitivity reactions<br>• Rare: pancreatitis, severe joint pain, SJS | • Caution with history of pancreatitis or pancreatic cancer<br>• CYP 3A4 interactions (except sitagliptin & alogliptin)<br>• Avoid in HF patients (saxagliptin) |

# PHARMACOLOGICAL MEASURES

## GLP-1 AGONISTS

| Drug class | MOA | Safety | C/I and D/I |
|---|---|---|---|
| **GLP-1 Agonists**<br>• Dulaglutide<br>• Liraglutide<br>• Lixisenatide<br>• Semaglutide<br><br>~0.6 – 1.4% | • ↑ incretin concentrations | AEs:<br>• N/V/D<br>• Injection site reactions<br>• Rare: acute pancreatitis, gallbladder disease<br>• Weight loss (1.1 – 4 kg) | • C/I: Pregnancy, personal/family hx of MTC or MENS2, caution with history of pancreatitis or pancreatic cancer<br>• ↓ gastric emptying (give agents requiring rapid GI absorption ≥ 1 hour before<br>• May need held prior to surgeries |

| GLP-1 AGONIST | Dosing Frequency |
|---|---|
| Liraglutide (with or without meals) | SC once daily |
| Dulaglutide (with or without meals) | SC weekly |
| Lixisenatide (prior to first meal of day) | SC once daily |
| Semaglutide (with or without meals) | SC weekly, PO daily |

# PHARMACOLOGICAL MEASURES

FYI

## GLP-1/GIP RECEPTOR AGONIST

| Drug class | MOA | Safety | C/I and D/I |
|---|---|---|---|
| **GLP-1/GIP Receptor Agonist**<br>Tirzepatide | • ↑ incretin concentrations | AEs:<br>• N/V/D<br>• Injection site reactions<br>• Rare: acute pancreatitis, gallbladder disease<br>• Weight loss<br>Other Safety:<br>• Insulin and sulfonylurea doses may need decreased when starting tizepatide<br>• Do not use with DPP4i or GLP1-RA | • C/I: Pregnancy, breastfeeding personal/family hx of MTC or MENS2, caution with history of pancreatitis or pancreatic cancer<br>• ↓ gastric emptying (give agents requiring rapid GI absorption ≥ 1 hour before)<br>• May need held prior to surgery |

Place in therapy is not well defined currently

# PHARMACOLOGICAL MEASURES

## SGLT2 INHIBITORS & INSULIN

| Drug class | MOA | Safety | C/I and D/I |
|---|---|---|---|
| **SGLT2 Inhibitors**<br>• Canagliflozin<br>• Empagliflozin<br>• Dapagliflozin<br><br>~0.5-0.7% | • Inhibits glucose reabsorption in the kidneys<br>• Empagliflozin canagliflozin and dapagliflozin slow nephropathy progression | AEs:<br>• Genital mycotic infection (e.g. yeast infection), hypotension, hyperkalemia<br>• Weight loss (2 – 3 kg)<br>• rare DKA<br>• ↑risk of fractures and lower extremity amputations with canagliflozin in one trial<br>• Reduced glycemic efficacy at lower eGFR | Renal Dosing (eGFR ml/min/1.73 $m^2$):<br>• Dapagliflozin – eGFR <25: do not start but may continue<br>• canagliflozin- eGFR <30: do not start but may continue<br>• Empagliflozin - eGFR <20: do not start but can continue<br>• Canagliflozin: Strong enzyme inducers (e.g. carbamazepine, rifampin, phenytoin)<br>• Loop diuretics ↑ risk of hypotension |
| **Insulin**<br><br>~0.9-1.2% or more | • Acts like endogenous insulin in the body | AEs:<br>• Weight gain, hypoglycemia, lipohypertrophy (must rotate sites) | • Oral antidiabetic drugs<br>• MAOI, βBs<br>• Salicylates<br>• Alcohol, Glucocorticosteroids |

# PHARMACOLOGICAL MEASURES

FYI

## ORLISTAT

| Drug class | MOA | Safety | C/I and D/I |
|---|---|---|---|
| **Lipase Inhibitor**<br>• Orlistat<br><br>(weight loss agent; may help reduce incidence of T2DM in at risk patients)<br><br>~0.2-0.4% | • Decreases fat breakdown and absorption by ~ 30%<br>• Minimal systemic absorption | AEs:<br>• Diarrhea, nausea, vomiting<br>• Increased bowel movements<br>• Flatulence<br>• Steatorrhea<br>• Abdo pain | • C/I in pregnancy, chronic malabsorption syndrome, cholestasis<br>• ↑ INR (warfarin)<br>• ↓ cyclosporine, levothyroxine (space 3-4 hours apart, respectively)<br>• ↓ absorption of fat-soluble vitamins (separate by 2 hours or more)<br>• ↓ anticonvulsants (monitor) |

# TYPES OF INSULIN

✓ Mixed insulins are also available (combination of rapid OR short-acting + intermediate-acting insulin)

| Insulin | RAPID-ACTING<br>Lispro (Humalog®, Admelog®)<br>Aspart (Novorapid®, Fiasp®)<br>Glulisine (Apidra®) | REGULAR or SHORT-ACTING<br>Humulin-R®<br>Novolin ge Toronto®<br>Entuzity 500 units/mL® |
|---|---|---|
| **Onset** | 10-15 mins (4 mins) | 30 mins (15 mins) |
| **Peak** | 60-120 mins (30-90 mins) | 2-3 h (4-8 h) |
| **Duration** | 3-5h (3-5 h) | 6.5 h (17-24 h) |
| **Basal/Bolus** | Bolus | |
| **Appearance** | Clear | |
| **Comments** | • Less hypoglycemia than regular insulin<br>• Take right before eating | • Take 30 min prior to eating |

# TYPES OF INSULIN

✓ Mixed insulins are also available (combination of rapid OR short-acting + intermediate-acting insulin)

| Insulin | **INTERMEDIATE ACTING**<br>*NPH*<br>Humulin N®<br>Novolin ge NPH® | **LONG-ACTING**<br>*Detemir*<br>Levemir® | **LONG-ACTING**<br>*Glargine*<br>Lantus®, Basaglar®<br>Toujeo® 300 units/mL | **ULTRA-LONG ACTING**<br>*Degludec*<br>Tresiba® |
|---|---|---|---|---|
| **Onset** | 1-3 h | 90 mins | | |
| **Peak** | 5-8 h | No pronounced peak | | |
| **Duration** | Up to 18 h | 6-24 h | 24h (> 30 h) | Up to 42 h |
| **Basal/Bolus** | Basal | | | |
| **Appearance** | Cloudy | Clear | | |
| **Comments** | • Given at bedtime or BID | • Peakless but detemir seems to have a shorter duration compared to glargine<br>• Usually daily dosing, detemir may need to be given BID at lower doses | | |

# TYPES OF INSULIN

FYI

Once weekly insulin was just released to the market in June 2024 and is not in guidelines yet.

| Insulin | **ONCE WEEKLY**<br>*Isulin icodec*<br>*Awiqli®* |
|---|---|
| **Onset** | Unknown |
| **Peak** | 15-18 h |
| **Duration** | Approximately one week |
| **Basal/Bolus** | Basal |
| **Appearance** | Clear and colourless solution |
| **Comments** | • Starting dose: 70 units once weekly OR previous basal dose (rounded to nearest 10) x 7<br>• Titrate by 20 units weekly (depending on patient factors) |

# PREMIXED INSULINS

FYI

| Type of Premixed Insulin | Trade Name/Components |
|---|---|
| Rapid-acting insulin + intermediate-acting insulin | Humalog® Mix 25 (25% lispro, 75% lispro protamine)<br>Humalog® Mix 50 (50% lispro, 50% lispro protamine)<br>Novomix® 30 (30% aspart, 75% aspart protamine) |
| Short-acting insulin + intermediate-acting insulin | Humulin® 30/70 (30% regular, 70% NPH)<br>Novolin® ge 30/70 (30% regular, 70% NPH) |

# SUMMARY

| Drug | HbA1c ↓ | FBG | PPG | Hypoglycemic Risk | Weight Gain |
|---|---|---|---|---|---|
| Metformin | ↓↓ | ↓ | ↓ | - | -/↓ |
| Sulfonylureas | ↓↓ | ↓ | ↓ | ↑ | ↑ |
| Meglitinides | ↓ - ↓↓ | ↓ | ↓↓ | -/↑ | -/↑ |
| TZDs | ↓↓ | ↓ | ↓ | - | ↑↑ |
| Acarbose | ↓ | ↓ | ↓↓ | - | -/↓ |
| DPP-4 Inhibitors | ↓ | ↓ | ↓↓ | - | -/↓ |
| GLP-1 Agonists | ↓ - ↓↓ | ↓ | ↓ - ↓↓ | - | ↓ |
| SGLT2 Inhibitors | ↓ | ↓ | ↓↓ | - | ↓ |
| Insulin | ↓↓↓ | ↓↓ | ↓↓ | ↑↑ | ↑↑ |

| | |
|---|---|
| Reduce A1C by 1-1.5% | Metformin, sulfonylureas, repaglinide, TZDs, GLP-1 analogues |
| Reduce A1C by ≤1% | Alpha-glucosidase inhibitors, DPP4 inhibitors, SGLT2 Inhibitors |

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# CHANGE LOG

**May 2025**

- Slide 2: Updated legend
- Slide 3: Updated eGFR C/I recommendation for metformin
- Slide 8: Removed UTI risk for SGLT2i and added 'genital mycotic infection.' Updated renal dosing info

**July 2025**

- No content changes made

**October 2025**

- No content changes made

**February 2026**

- Slide 8: Renal dosing updated